

*MA.8.AR.3.3***Benchmark**

MA.8.AR.3.3 **Given a table, graph or written description of a linear relationship, write an equation in slope-intercept form.**

Connecting Benchmarks/Horizontal Alignment

- MA.7.AR.4.4
- MA.8.AR.3.2, MA.8.AR.3.5
- MA.8.F.1.1
- MA.8.DP.1.3

Terms from the K-12 Glossary

- Intercept
- Slope

Vertical Alignment**Previous Benchmarks**

- MA.6.AR.4.4
- MA.7.AR.4.4

Next Benchmarks

- MA.912.AR.2.2, MA.912.AR.2.3

Purpose and Instructional Strategies

In previous courses, students determined rates, and unit rates in ratios. In grade 7 accelerated, students translate between different representations of proportional relationships. Students also write an equation in slope-intercept form from a written description, a table, or a graph. In Algebra 1, students will write a linear two-variable equation to represent a relationship given by a variety of mathematical and real-world contexts.

- Point-slope form and standard form are not expectations at this grade level.
- Instruction connects proportional relationships to support the generation of the equation $y = mx + b$. Helping students see how the linear equation is both the same and different from the proportional relationship will support the appropriate use of proportional thinking using the rate of change.
- Using an online dynamic graphing tool to explore how the graph changes as either the slope or the y -intercept changes help students visualize the coefficients and constants in the equation (*MTR.4.1*).
- Students should recognize in a table that the y -intercept is the y -value when x is equal to 0. The slope can be determined by finding the ratio between the change in two y -values and the change between the two corresponding x -values.
- Using graphs, students identify the y -intercept as the point where the line crosses the y -axis and the slope as the vertical change divided by the horizontal change. In a linear equation, the coefficient of x is the slope and the constant is the y -intercept. Students need to be given the equations in formats other than $y = mx + b$, such as $y = ax + b$ or $y = b + mx$.
- Instruction includes using a variety of vocabulary to make connections to real-world concepts and future courses. To describe the slope, one can say either “the vertical change divided by the horizontal change” or “rise over run.”

- The instruction includes examples where the slope is positive or negative and the y-intercept is given as a positive or a negative in the equation.
- When providing a graph, be sure there are easily identifiable points for students to use in calculating the slope.
- Instruction allows students to make connections between the different representations of a linear relationship (*MTR.2.1*).

Common Misconceptions or Errors

- Students may incorrectly identify the values for the slope and y-intercept.
- Students may incorrectly calculate the slope with a common error of inverting the change in y and the change in x.

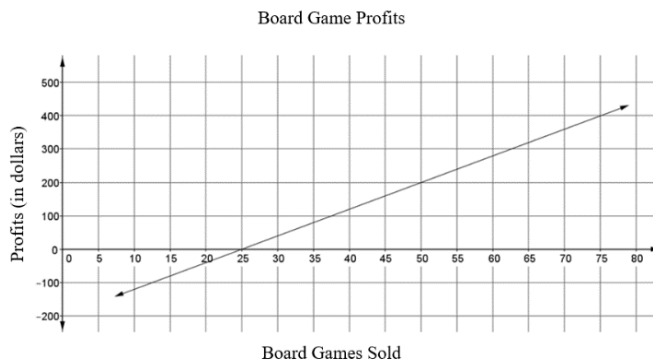
Strategies to Support Tiered Instruction

- Teacher supports students who incorrectly identify the values for the slope and y-intercept by providing opportunities to notice patterns between a given value for b , a line graphed on the coordinate plane, and a given equation of the same line.
- Teacher supports students who incorrectly calculate the slope by inverting the change in y and the change in x using error analysis tasks, in which the expression $\frac{y_2 - y_1}{x_2 - x_1}$ is incorrectly written as $\frac{x_2 - x_1}{y_2 - y_1}$, and has students find and correct the error.
- Teacher supports students who invert the x and y values when calculating slope by using real-world problems that students can relate to and helping students represent the relationship visually.
- Teacher co-creates an anchor chart naming the slope and y-intercept of a given line and then discusses where to start when graphing the line.
- Instruction includes graphing various linear equations from a table and then discussing the pattern students notice in regard to the y-intercept.
- Teacher provides students with graphs and equations of several linear equations then co-illustrates connections between the slopes and y-intercepts of each line to the corresponding parts of each equation using the same color highlights.
- Teacher co-creates a graphic organizer with students to include examples of positive and negative slope; the meaning of each variable in slope intercept form; and how to determine the slope and y-intercept in a table, graph and verbal description.

Instructional Tasks

Instructional Task 1 (MTR.7.1)

Victoria owns a store that sells board games. She made the following graph to relate the number of board games she sells to her overall profits.



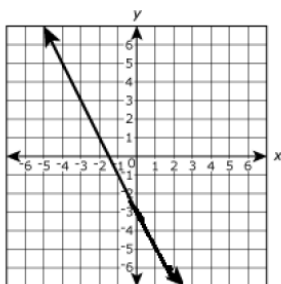
Part A. Write an equation in slope-intercept form to describe this relation. Explain how to determine the equation.

Part B. What is the meaning of the y-intercept in the given context? What is the meaning of the slope in the given context?

Instructional Items

Instructional Item 1

Write an equation that represents the graph shown.



Instructional Item 2

The table shown represents a linear relationship. Using the table, write an equation in slope-intercept form.

x	y
0	-4
1	-1
2	2
3	5

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*